A palaeomagnetic investigation of the Ulvö dolerite, Ångermanland, central Sweden

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A palaeomagnetic study of dolerites spread over a large area in the coastal part of Ångermanland, central Sweden, introduces a palaeopole close to poles obtained from Jotnian or post-Jotnian dolerites in Finland, implying that all these rocks belong to the same intrusive phase.

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The purpose of this study was to decide if dolerites of the Ulvö type, spread over a large area, were approximately of the same age.

The investigated Ulvö dolerites are of post-Jotnian age because they intrude Jotnian sedimentary rocks (seen in the Nordingrå district). Radiometric dating of the dolerites in Ångermanland (at Långroudden, Näsänget and Häggdånger) indicates a K-Ar age of about 1245 ± 20 m.y. (Welin & Lundqvist 1975).

The dolerite province is located along the coast of Ångermanland, but dolerites also outcrop as a semi-circle further into the county, (Fig. 1).

The dolerites intrude in a sheet-like manner with a gentle dip and with a thickness of about 250 metres. However, gravity studies (Larson & Magnusson, in prep.), have shown the dolerites in some areas to measure up to 800 metres in thickness and to be of lopolithic shape. Another characteristic of the dolerites is the rhythmic igneous layering (Larson 1973).

In several areas within the dolerite province, faults can be traced as lineaments seen on topographic maps. By calculations of gravity data these faults measure up to 500 metres (Larson & Magnusson, in prep.).

Mineralogy

The Ulvö dolerite shows a simple mineralogy. The main minerals are plagioclase (labrador-bytownite), augitic pyroxene, olivine, biotite and Fe-Ti oxides. Accessories can be apatite, potash feldspar, sericite, serpentine, amphibole, epidote and quartz.

The Fe-Ti oxides consist of magnetite, ulvite and ilmenite. The grains are usually complex and commonly contain all three phases. Ulvite forms an intergrowth with magnetite in a discrete network, dividing the host magnetite into small areas. Ilmenite usually takes the form of lamellar intergrowths along the octahedral planes of the magnetite. It may also occur as separate ilmenite grains, patchy ilmenite in magnetite or as separated areas in the magnetite.

The ulvite phase contains 31–35 mol.% Fe_3O_4 in solid solution and the magnetite

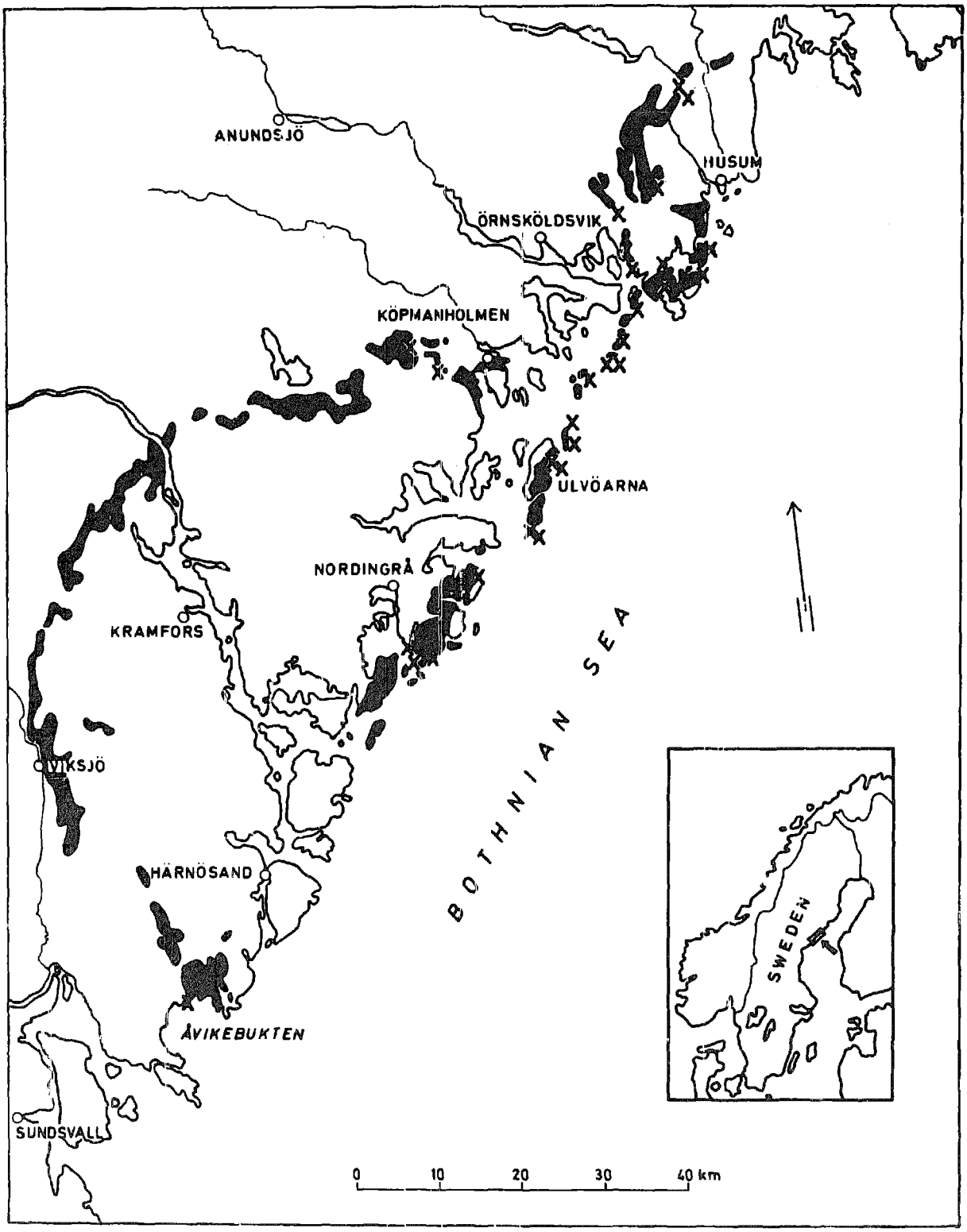


Fig. 1. Map showing the location of the investigated area and sample localities (x). Black is dolerite. (Modified from Lundqvist & Samuelsson 1973.)

phase contains about 0–13 mol.% Fe_2TiO_4 in solid solution (Larson & Magnusson 1976).

Sampling and measurement

The sampling covers the main part of the Ulvö dolerite area (Fig. 1). All samples were hand-picked and for orientation a magnetic compass as well as a tripod was used. In all 32 large hand samples were demagnetized and measured with an Oerstedt meter.

Each sample was AC-demagnetized using peak fields of 100, 200, 300 and 500 Oe. Three samples still had about 1/5 of their remanence intensity left after demagnetization in a 500 Oe peak field (Fig. 2). One of these samples was demagnetized in peak fields of up to 2100 Oe in steps of 200 Oe. The sample shown in Fig. 2a is influenced by a viscous component; the other two samples (b and c) do not reveal any viscous component but seem to be composed of a single characteristic magnetization. However, it must be emphasized that the sample showing the strong effect of viscous magnetization is the most extreme case of this effect among the samples studied.

Other samples have less than 1/10 of their natural remanent magnetization (NRM) intensity left after 500 Oe treatment. The resistance to AC-demagnetization suggests that the remanent magnetization is associated with single domain to pseudo-single domain grains (Larson & Magnusson 1976).

The densest distribution of RM directions was found to be at 200 Oe peak field demagnetization (Fig. 4). After 300 Oe the directions start to spread (cf. Table 1). This is due to the small proportion of remanent magnetization left above this peak field. The distribution of NRM and RM directions is shown in Figs. 3, 4, 5 and 6.

At 200 Oe treatment a diffuse trend is visible in a NNE–SSW direction. However, at higher peak fields this trend is not significant. If the diffuse trend is caused by a linear com-

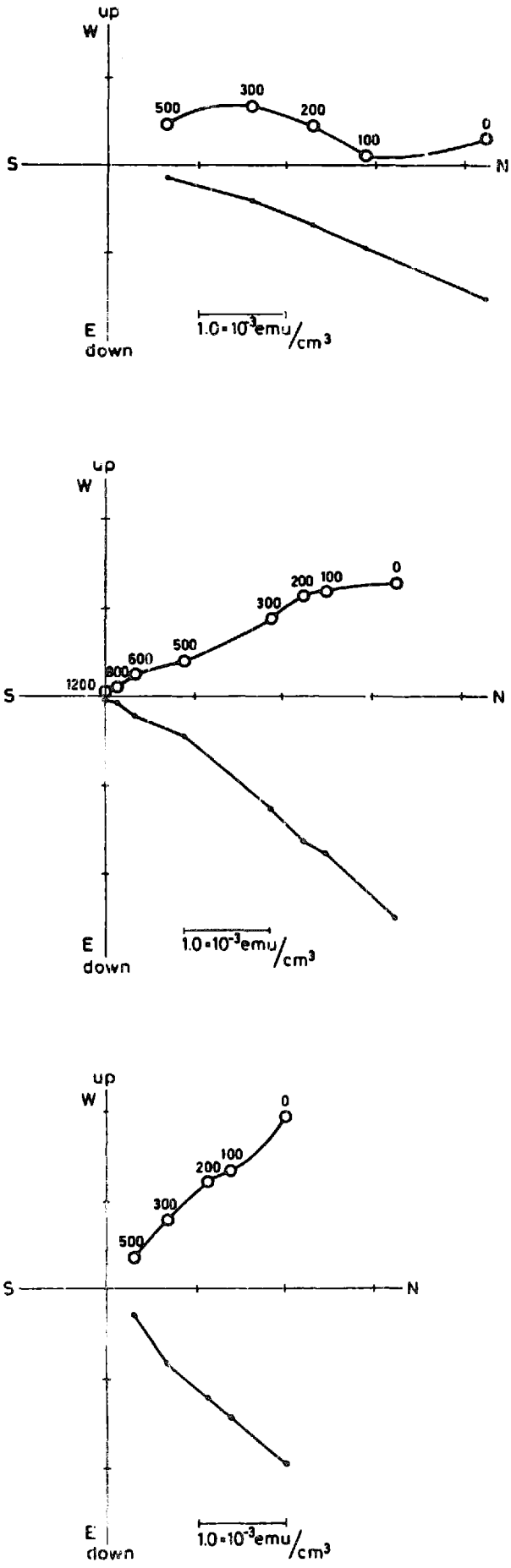


Fig. 2. Demagnetization diagrams of three samples (discussed in the text). (a) shows a viscous magnetization effect. (b) and (c) show a characteristic magnetization. Numbers denote different demagnetization steps (Oe). Open circle: projection of the RM-vector on the horizontal plane. Dot: projection of the RM-vector on the vertical N–S plane.

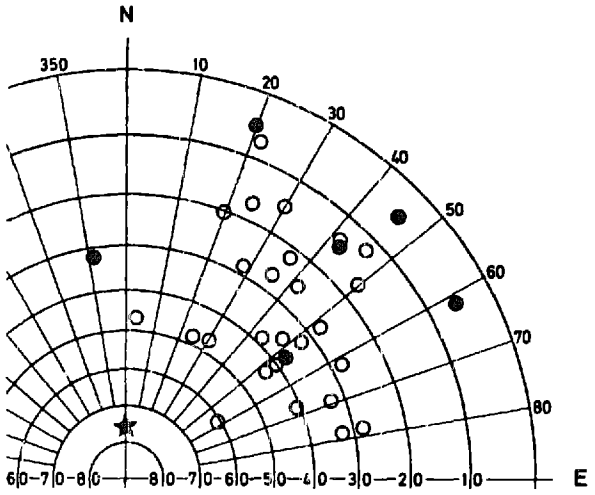


Fig. 3. Directions of natural remanent magnetization plotted in stereographic projection. Open circle: Negative inclination. Filled circle: Positive inclination. Star: Present geomagnetic field.

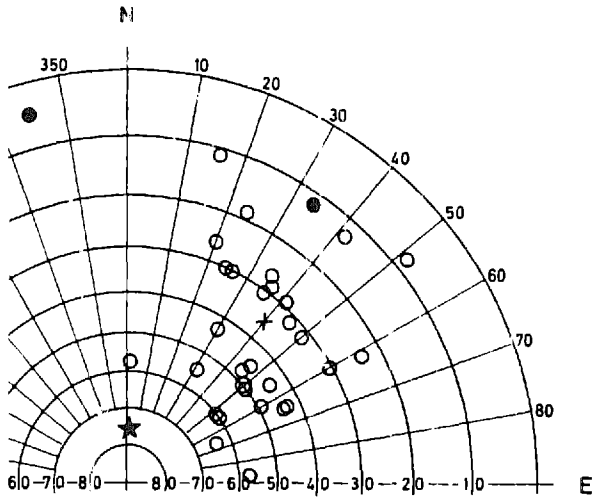


Fig. 4. Directions of remanent magnetization after AC-demagnetization 200 Oe. Cross: Mean direction of the samples.

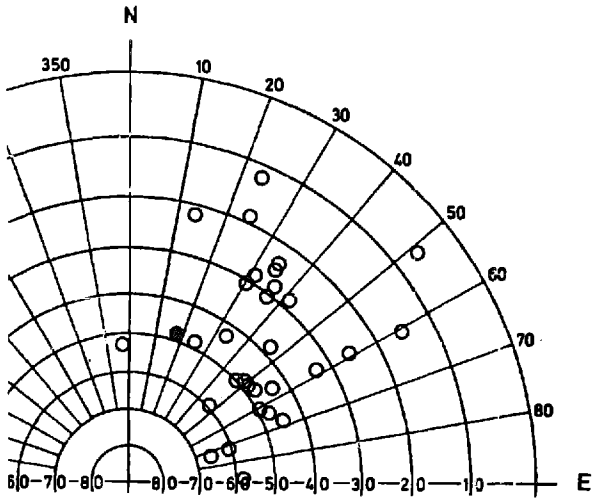


Fig. 5. Directions of remanent magnetization after 300 Oe. Symbols as in Fig. 3.

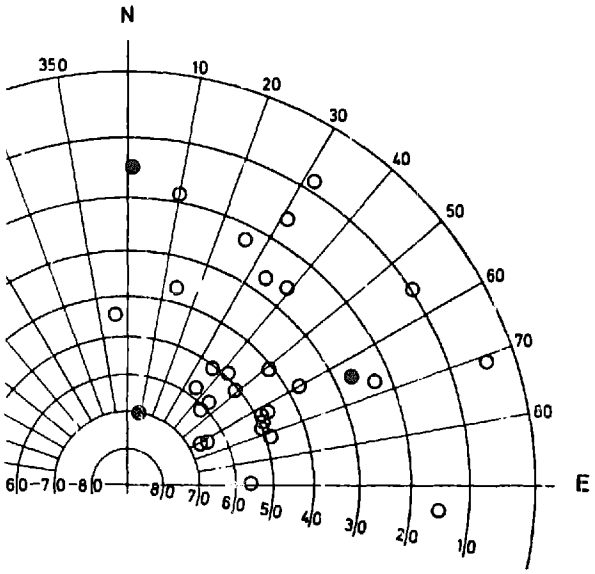


Fig. 6. Directions of remanent magnetization after 500 Oe. Symbols as in Fig. 3.

bination of two poles they are probably close. It is not possible to relate directions from extremes of the trend to specific geographic areas. In earlier works on the Jotnian palaeopole in the surrounding dolerite areas of the Baltic shield (Neuvonen 1965, 1966; Neuvonen & Grundström 1969; Larson & Magnusson 1976), this trend has not been observed. However, in these investigations the collected samples have been chosen from more restricted areas and have given a smaller spread in directions. The trend observed could be due to a gentle tilting of blocks in a regular way. This could be a reasonable suggestion as the sampling area measures about 80 km in a north–south direction (Fig. 1).

The investigation of the magnetic mineral phases (Larson & Magnusson 1976) has not revealed any sign of secondary phases connected to an event later than the dolerite intrusion. All sampled rocks have been very fresh.

Table 1. Direction of remanent magnetization after AC-demagnetization using Fischer's statistics.

Number of samples measured	Peak demagnetization field (Oe)	Declination (°) mean direction	Inclination (°)	α95 (°)	k
32	NRM	45.8	−28.1	20.2	2.6
32	100	41.9	−31.4	8.0	11.0
32	200	40.0	−35.8	7.9	11.3
32	300	40.0	−37.2	9.5	8.2
32	500	43.7	−35.3	13.5	4.6

α95=half-angle of cone of confidence, k=precision parameter.

The viscous magnetization tends to oppose a negative inclination and also gives a smaller declination, resulting in an approximate trend as that observed. However, the influence on the inclination will be most pronounced. According to this, Fischer's statistics (Fischer 1953) will give a Jotnian pole position somewhat displaced, but not seriously. We therefore suggest that Fischer's statistics give a reasonable estimate of the statistical parameters. The directions and statistical parameters at different demagnetization steps are listed in Table 1.

Palaeomagnetic pole position

The directions of the natural remanent magnetization are rather concentrated (Fig. 3) and different from the present geomagnetic field. This fact indicates favourable conditions for obtaining a palaeomagnetic direction.

The inferred palaeomagnetic pole position of the present study and a previous result from the Ulvö dolerite area along with other relevant palaeomagnetic poles from the Baltic shield are listed in Table 2. The palaeomagnetic pole position of the present study is plotted together with the polar wandering curve for Europe (Fig. 7). The pole position of the Ulvö dolerite (U), 1°N, 161°E, fits well with the constructed curve and is situated between the poles of the Satakunta (SD) and Märket (M) dolerites (Neuvonen 1966; Neuvonen & Grundström 1969).

Conclusions

Especially three samples were found to have a hard remanent magnetization, due to a fine intergrowth of ulvite and ilmenite dividing the host magnetite into small areas (Larson & Magnusson 1977). The formation of this texture is closely related to the time of intrusion of the dolerite. Because of the good agreement between the palaeomagnetic pole obtained by Neuvonen (1965, 1966) and Neuvonen & Grundström (1969) and the present study, it can be stated that no major block rotation has occurred between the Ulvö dolerite province and the Finnish dolerites since Jotnian time. This investigation also suggests that dolerites from different areas in the Ulvö dolerite province are intimately connected in time.

Recently another pole position from the Ulvö dolerite has become known to us (Poorter 1976). Though based on samples from two sites only (Nordingrå) the palaeomagnetic pole

Table 2. Pole positions of rocks in Finland, Norway and Sweden with an age approximately the same as that of the Ulvö dolerite.

Rock unit and country	Pole position Long. (°)	Lat. (°)	Age	Reference
Satakunta dolerites, Finland	158 E	2 N	970–1330 m.y., K/Ar	Neuvonen 1965
Vaasa dolerite, Finland	164 E	7 N	Jotnian	Neuvonen 1966
Märket dolerite, Finland	146 E	6 S	Younger than Rapakivi	Neuvonen & Grundström 1969
Egersund dikes, Norway	129 W	22 S	850–950 m.y.	Poorter 1972
Jotnian basalts, Sweden	174 W	32 N	745–931 m.y., K/Ar	Priem et al. 1968 and Mulder 1971
Late Jotnian dolerites, Sweden	178 E	23 N	Late Jotnian	Priem et al. 1968 and Mulder 1971
Hyperite dolerites, Sweden	134 W	12 S	781–1573 m.y., K/Ar	Priem et al. 1968 and Mulder 1971
Dolerites and basalts, Sweden	115 E	10 S	1200 m.y., K/Ar	Dyrelius 1970
Ulvö dolerite, Sweden	153 E	4 S	1245 m.y., K/Ar	Larson & Magnusson 1976 and Welin & Lundqvist 1975
Ulvö dolerite, Sweden	161 E	1 N	1245 m.y., K/Ar	Present investigation

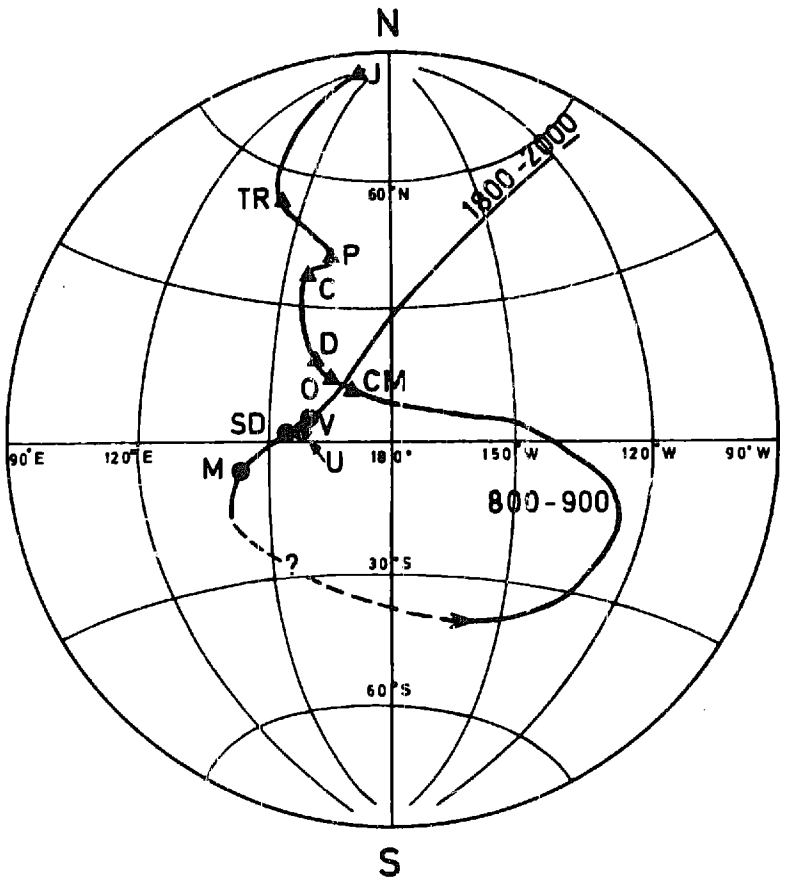


Fig. 7. Suggested polar wander-curve for Europe (Spall 1975 and Creer 1970). The palaeomagnetic pole (U) of the Ulvö dolerite is pointed out by an arrow. M: Märket dolerite. V: Vaasa dolerite. SD: Satakunta dolerite (Neuvonen 1965, 1966; Neuvonen et al. 1969). Abbreviations CM, O, D, C, P, TR and J indicate mean pole positions of Cambrian to Jurassic age.

(158°E, 5°S) is in very good agreement with that of the present paper. Poorter also reports that the viscous component was cleaned after 200–300 Oe treatment.

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